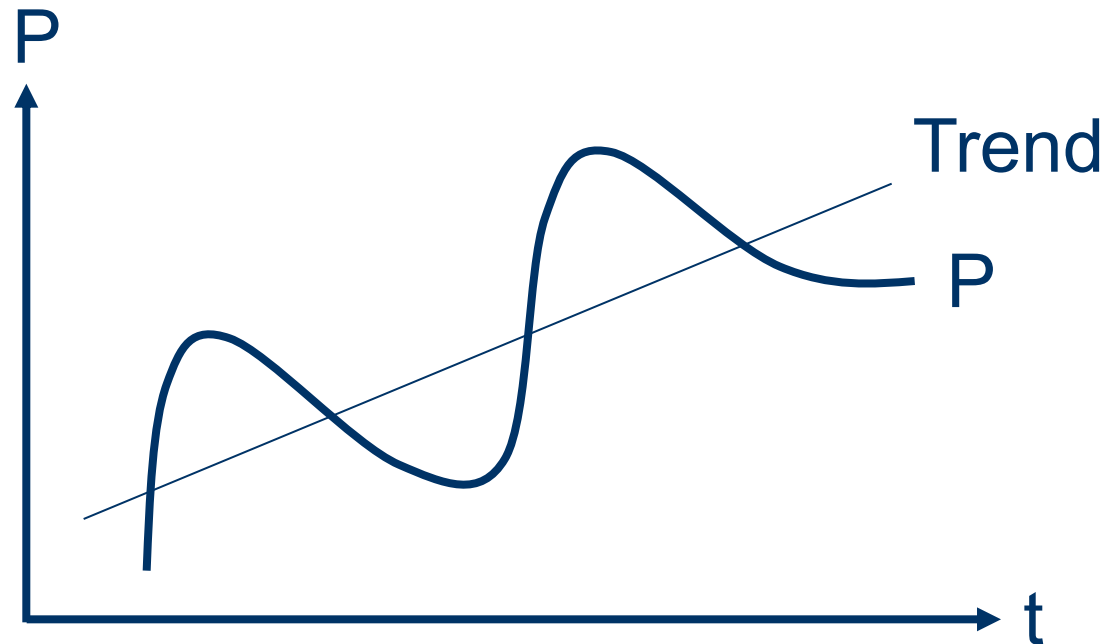


Topics 4/5/6—Seminar 5 (WT Week 9)
**Property Market Dynamics and
Empirical Tests**



Contents—*Seminar 5*

1. Group presentations & discussion
2. How to approach an exam / exam question?
 - GY457 exam specifics
 - Nature of exam questions
 - How to answer exam questions?
 - This question (Q7 from 2012/13 exam)

⇒ How to Approach an Exam / Exam Question...?

GY457 Exam Specifics

- Two parts – 4 questions
 - ▶ Part 1 consists of 2 questions (Bagagli lectures)
 - ▶ Part 2 consists of 2 questions (Hilber lectures, Yu lecture)
 - ▶ Questions may overlap topics (but not Parts 1 and 2)
- Answer 2 questions – one from each part
- Time allowed: 3 hours

Nature of Exam Questions

- Usually focused on one or a few topics, *for example*:
 - ▶ Tiebout-fiscal decentralization-capitalization (just Topic 1)
 - ▶ Real estate cycles (Topics 4-6)
 - ▶ Homeownership (just Topic 7)
 - ▶ Land use regulation & real estate cycles (Topic 2 and Topics 4-6)
 - ▶ *And including* Housing issues in China (just Topic 8)
 - Can draw on several blocks (within the term)
 - ▶ Can e.g., overlap two or three topics in Blocks I, II and III
- ⇒ Strong advice to prepare whole syllabus!

Nature of Exam Questions—*cont.*

- Questions always draw on relevant material but can be ‘unexpected’ (not covered specifically in lectures and seminars)
 - ▶ Example: In AT, effect of government forbidding intensive food production on urban form
- Always ‘new’ questions drawing from same material!

How to Answer Exam Questions?

- Distinguish between open (essay-style) questions and questions with partitions
 - ▶ Each part will have one open (essay-style) question and one question with partitions
- **Open (essay-style) questions**
 - ▶ Structure your answer (perhaps sketch)
 - ▶ Essay should have intro, main body (analysis) & conclusion/synthesis
 - ▶ Give answer that reflects all the main aspects of the question (but nothing else!)

How to Answer Exam Questions?—*cont.*

- **Partitioned questions**

- ▶ Answer sub-question ‘to the point’
- ▶ Don’t lose time with answering one sub-question in excessive detail leaving no time for the rest
 - Equal weight to sub-questions unless otherwise noted

Real Estate Cycle Question: Part a)

- Discuss two theories (not more/not less!)
- Good response discusses in some depth (i) main mechanism, (ii) predictions, (iii) supporting evidence of two theories
- Good answer adds ‘flesh’ to lecture notes from various articles on reading list

Real Estate Cycle Question: **Part a)**— *cont.*

- Possible theories to discuss
 - 1) Inelastic long-run supply & local business cycles (**Hilber & Vermeulen 2016; Hilber & Mense 2025**)
 - 2) Irrational exuberance (**Case et al. 2012**; Shiller 2014; Case & Shiller 1998, 1989, 2003)
 - 3) Myopic agents & lags (**Wheaton 1999**)
 - 4) Liquidity constraints (Stein 1995; **Ortalo-Magne & Rady 2006**; Lamont & Stein 1999; **Genesove & Mayer 1997**)
 - 5) Real option theory (**Grenadier 1995 & 1996**)
 - 6) Search and matching (**Head et al. 2014**)
 - 7) *Loss aversion [does not explain full cycle]* (Kahneman & Tversky 1979; Tversky & Kahneman 1991; **Genesove and Mayer 2001**)

Real Estate Cycle Question: Part b)

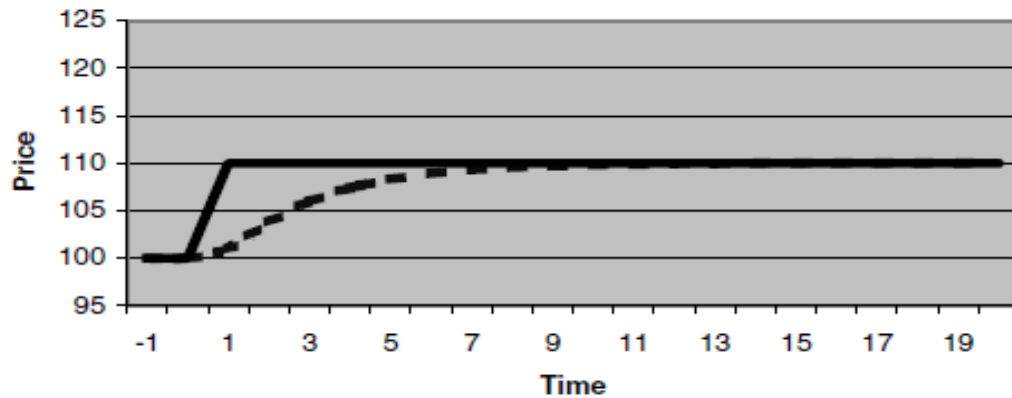
- The standard method = mean-reversion framework / error-correction model
 - i. Regress HP on D & S fundamentals → Predict ‘fundamental’ HP
 - ii. Deviation of actual HP from ‘fundamental HP’ can be interpreted as ‘overshooting’ or ‘undershooting’ (due to e.g., irrational exuberance, myopic agents & lags)
 - iii. Estimate error-correction model with serial correlation coefficient (α), mean reversion coefficient (β) and immediate adjustment to ‘fundamental price’ (γ)
$$\Delta P_t = \alpha \Delta P_{t-1} + \beta (P_{t-1}^* - P_{t-1}) + \gamma \Delta P_t^*$$
 - iv. Markets are cyclical if $4\alpha > (1 + \alpha - \beta)^2$ (price adjusts to shock in oscillations)

Real Estate Cycle Question: **Part b)**—cont.

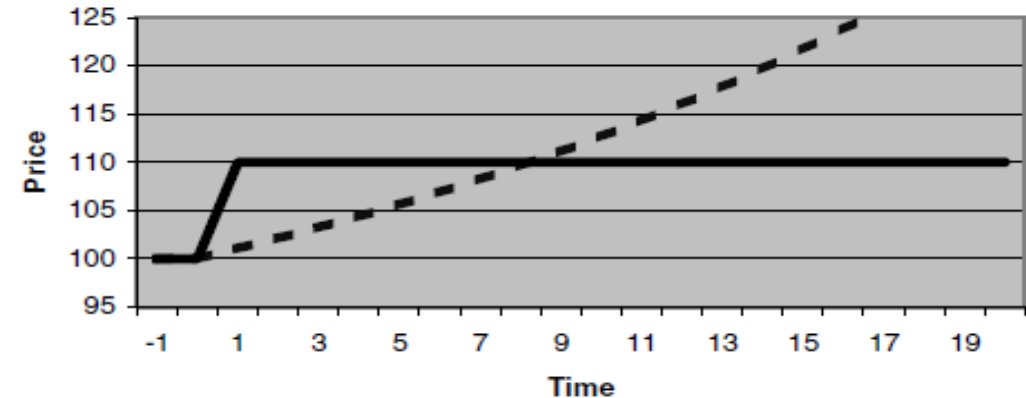
$$4\alpha < (1 + \alpha - \beta)^2$$

$$4\alpha > (1 + \alpha - \beta)^2$$

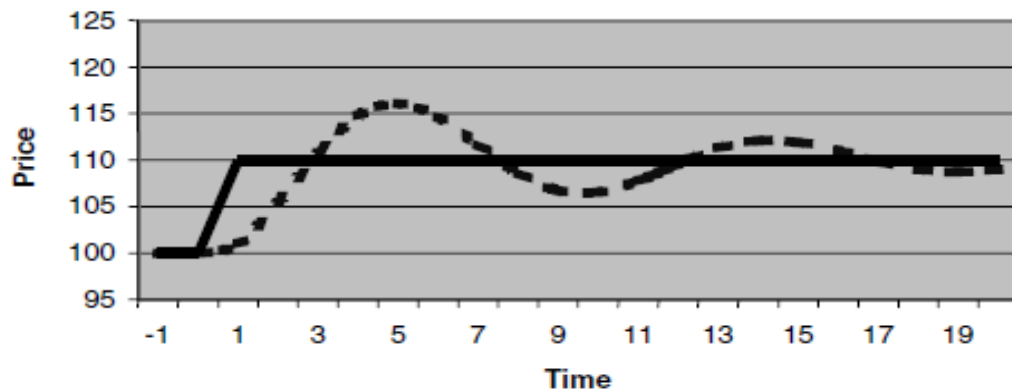
Region I: Convergent, No Oscillations



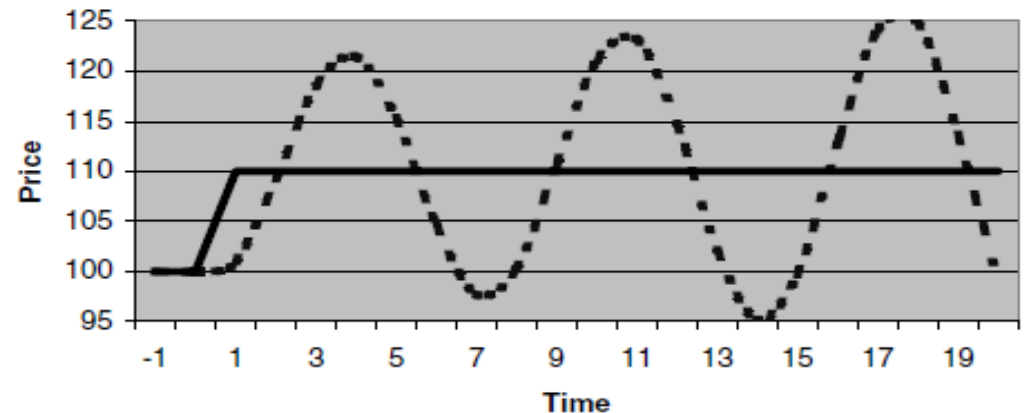
Region II: Divergent, No Oscillations



Region III: Convergent Oscillations



Regions IV: Divergent Oscillations



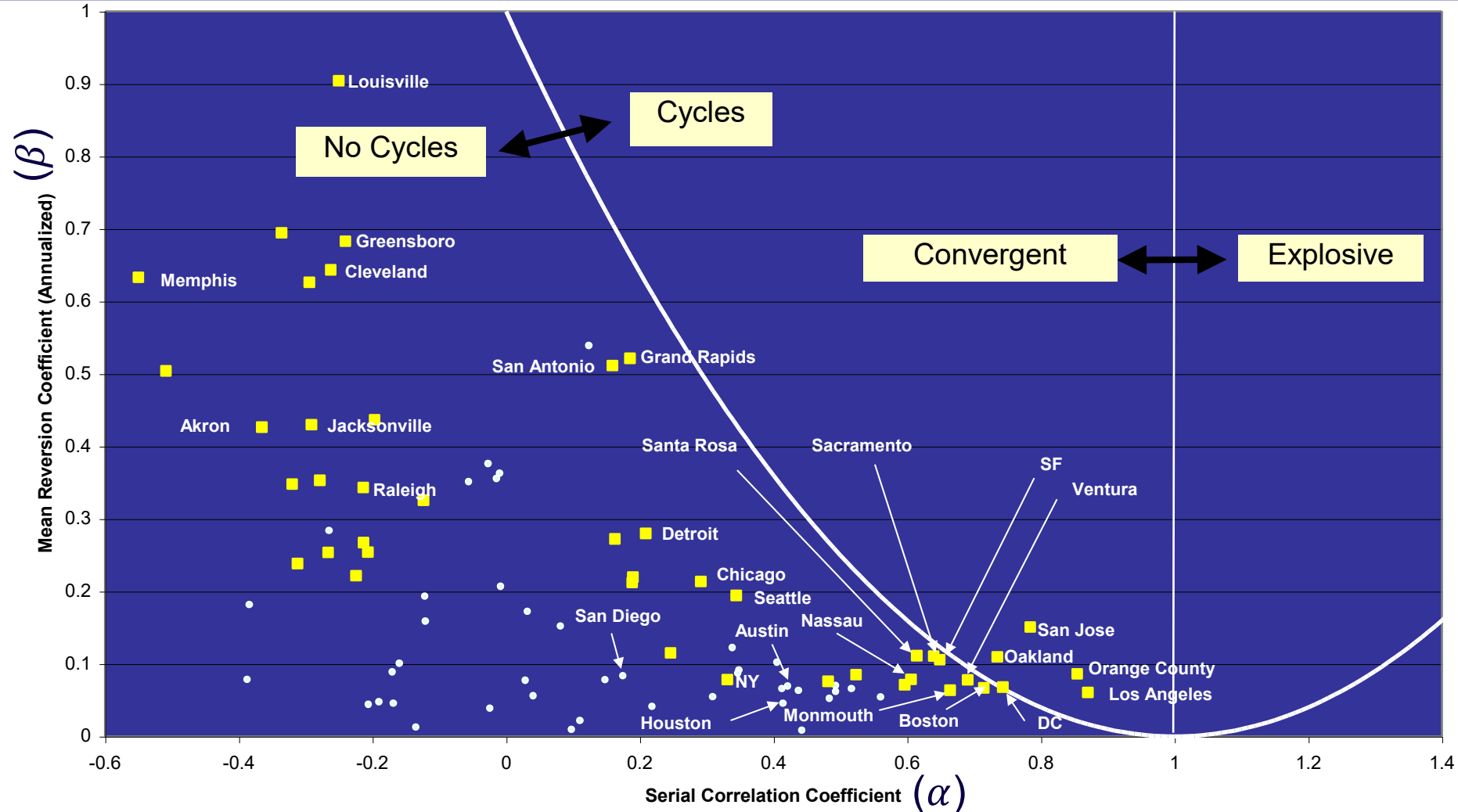
$$\alpha < 1$$

$$\alpha > 1$$

α ...Serial correlation coefficient
 β ...Mean reversion coefficient

Real Estate Cycle Question: **Part b)**—cont.

Relationship b/w Serial Correlation & Mean Reversion for U.S. Metro Areas (Pooled Regression, 1980-2001)



Source: Hilber, C. (2003)

Real Estate Cycle Question: **Part b)**—*cont.*

- Main empirical difficulty = Estimation of ‘fundamental price’
 - ▶ Suffers from omitted variables (OVs) problem (unobserved and unobservables)
 - ▶ Correct functional form of ‘fundamental equation’?
 - ▶ Is residual really capturing disequilibrium phenomena or just the result of e.g., OV-problems?

Real Estate Cycle Question: Part c)

- More open-ended question – not easy to answer
 - ▶ Differentiates b/w good and top answers
- Several possible explanations
 - ▶ Not driven by long-run supply constraints but...

Real Estate Cycle Question: **Part c)**—cont.

- Possible explanations
 - ▶ Short-run supply constraints (e.g., construction lags) in conjunction with positive initial demand shock, myopic agents & long-run elastic supply → endogenous oscillations (**Wheaton 1999**)
 - ▶ Short-run supply constraints & **contagion** with irrational exuberance (**DeFusco et al. 2013**)
 - ▶ Undeveloped land is a pure frictionless bet on real estate → plenty of undeveloped land & disagreement about market fundamentals (long-run supply constraints in LV soon binding?) provide market for speculation (**Nathanson & Zwick 2013**)
 - ▶ Others?

Note on measuring cyclicity

- Mathematical methods allow us to derive a “characteristic equation” for our estimated model equation. The characteristic equation shows that markets have a cyclical behavior if $4\alpha > (1 + \alpha - \beta)^2$.
- In our **cycle rating** we defined a MSA as a **cyclical** market if it fulfills this condition. If a MSA has a non-cyclical behavior, that is, $4\alpha < (1 + \alpha - \beta)^2$, we define the MSA as **non-cyclical** market.
- As the key parameters are always estimated with errors, we define markets as **near cyclical** if $0 > 4\alpha - (1 + \alpha - \beta)^2 > -0.2$.

Source: Hilber, C. (2003)

Note on ‘endogenous cycles’ vs. ‘exogenous cycles’

- Error correction model equation:

$$\Delta P_t = \alpha \Delta P_{t-1} + \beta (P_t^* - P_{t-1}) + \gamma \Delta P_t^*$$

- If $\gamma > 0$: Business cycles cause cyclical response; the higher γ , the stronger the response [serial correlation & mean reversion *in business cycle* $\Rightarrow$ **exogenous cycle**] [$\gamma=0$ would mean fundamental adjustments are irrelevant; γ is amplifier]
- If $4\alpha > (1 + \alpha - \beta)^2$: One exogenous shock triggers convergent oscillations towards new equilibrium $\Rightarrow$ **endogenous cycle**]